



DEPARTMENT OF MATHEMATICS AND COMPUTING
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
End Semester Examinations
B. Tech MC A IV Semester
MAPC 204: Computer oriented numerical methods

Date: 09-08-2026

Time: 02.00 PM - 05.00 PM

Max. Marks: 50

Answer the entire question at one place. State all the theorems and results used.

Q1. For the equation $f(x) = x^2 - 3x + 2 = 0$, each of the following functions yields an equivalent fixed-point problem: [8]

(a) $g_1(x) = \frac{x^2 + 2}{3}$

(b) $g_2(x) = 3 - \frac{2}{x}$

(c) $g_3(x) = \frac{x^2 - 2}{2x - 3}$

Analyze the convergence properties of each of the corresponding fixed-point schemes for the root $x = 2$. Confirm your analysis by computing atleast 3 iterations with $x_0 = 3.0$ and verify the convergence (or lack thereof). Among the three fixed-point iterations, identify the iteration that converges fastest to the root $x = 2$. Determine its order of convergence and justify your answer.

Q2. The cross-sectional area of a storage tank was measured at different depths as given below: [8]

Depth, h (m)	0	1	2	3	4	5	6
Area, $A(h)$ (m^2)	12.0	13.5	15.8	18.2	20.1	21.4	22.0

The volume of the tank up to a depth of 6m is given by

$$V = \int_0^6 A(h) dh.$$

Estimate the volume using the composite trapezoidal rule, and Simpson's 3/8 rule. Compare the two estimates and explain the difference between them.

Q3. A drug is administered into the bloodstream, and its concentration $C(t)$ is modeled by the initial value problem [8]

$$\frac{dC}{dt} = -0.15C + 5e^{-0.1t}, \quad C(0) = 20.$$

Calculate the concentration of the drug at $t = 0.5$ and $t = 1.0$, using the Euler's method and the second-order Runge-Kutta method. Consider $h = 0.5$.

Q4. [8]

(a) In the following difference table, determine all the missing entries (marked with ?), using the relations between $f(x)$ and its forward differences.

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$
0	3	?	2
1	?	5	?
2	11	?	
3	18		

(b) The sum of two numbers is 20. If each number is added to its square root, the product of the two sums is 155.55. Determine the two numbers to within 10^{-3} .

Q5. Three different types of fertilizers contain nitrogen, phosphorus, and potassium in the proportions shown below: [6]

Fertilizer	Nitrogen	Phosphorus	Potassium
A	10	5	2
B	6	10	4
C	3	3	10

A farmer requires a mixture containing 120 units of nitrogen, 100 units of phosphorus, and 110 units of potassium.

- Formulate the corresponding system of linear equations.
- Solve the system using the Gauss-Siedel iterative method, starting with

$$x^{(0)} = y^{(0)} = z^{(0)} = 0.$$

Perform three iterations and determine the amount of each fertilizer to be used in the mixture.

Q6. Determine the dominant eigenvalue and the corresponding eigenvector of

[6]

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Start with $X^{(0)} = [1 \ 1 \ 1]^T$ and perform at least three iterations.

Q7. The following table gives the amount of dissolved oxygen in a water sample:

[6]

Time, t (h)	0	2	4	8	10
Dissolved oxygen, $D(t)$ (mg/L)	8.4	7.2	6.1	4.0	3.2

The measurement at $t = 6$ hours was lost. Use an appropriate interpolation technique to estimate the dissolved oxygen level at $t = 6$ hours. Explain why the chosen interpolation technique is suitable for recovering the missing measurement.